

Week 11 Lab: Skill Wrap-Up

AAI1001 – Data Engineering and Visualization

AY25/26 Tri 3 (Assisted by GenAI)

Assessment 3 (Teamwork 2)

AUTHOR

Guang Hua

AFFILIATION

Singapore Institute of Technology

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Lab Team-Based Assessment Rules

Academic integrity is of utmost importance. Therefore, all solutions submitted must be the result of your team's sole collaborative effort during the allocated two-hour lab session. To maintain academic integrity, please follow these guidelines:

1. Teams may access the World Wide Web through their laptops, and the use of generative artificial intelligence (GenAI) is permitted.
2. Printed materials are allowed on the desk.
3. **Avoid real-time communication with anyone outside your team, except the instructors.** Forbidden communication channels include, but are not limited to, email, messenger apps, and shared online accounts. Do not engage in discussions with members of other teams and refrain from viewing their monitors.
4. **Exchanging files with anyone** (e.g., using [GitHub](#), [Dropbox](#), or [RStudio Server Pro](#)) **other than your teammates and the instructors is strictly forbidden**, whether before, during, or after the lab exercise. This rule extends to shared code that could potentially be used by other teams, including those in future iterations of this course.

Non-compliance with these rules will lead to appropriate disciplinary procedures.

Introduction

This team-based assessment tests your ability to work collaboratively on data engineering and visualization tasks using the [tidyverse](#) suite, [gt](#) package, and other analytical tools. As a team, you are required to collaboratively write a file, `team_XX.qmd` (replace `XX` with your team number) and render a single-page [HTML](#) report with text, code, and results. You are expected to set proper formatting, define an appropriate document structure, and demonstrate clear and coherent data workflow. Before you start, apply the following basic settings to your `team_XX.qmd` file:

- **Embed resources** to minimize the files needed for submission.
- **Add section and subsection numbers** for easy reading.
- Set proper code execution options to **display ALL code and output**.
- Use `|>`, **NOT** `%>%`, for piping in your code chunks.
- For this assessment, GitHub is **NOT** compulsory.

Task 1: English Premier League Table 2019–2020 (20%)

Download the file `epl.zip` from xSITe (**Content**→**Data**). The file contains:

1. `epl_teams.csv`: Names and three-letter abbreviations of all teams in the English Premier League (EPL) during the 2019–2020 season.
2. `epl_results.csv`: Results of all EPL matches during the 2019–2020 season, shown below.

Content in <code>epl_results.csv</code>																				
home_team	ars	avl	bou	bha	bur	che	cry	eve	lei	liv	mci	mun	new	nor	shu	sou	tot	wat	whu	wol
ars	-	3-2	1-0	1-2	2-1	1-2	2-2	3-2	1-1	2-1	0-3	2-0	4-0	4-0	1-1	2-2	2-2	3-2	1-0	1-1
avl	1-0	-	1-2	2-1	2-2	1-2	2-0	2-0	1-4	1-2	1-6	0-3	2-0	1-0	0-0	1-3	2-3	2-1	0-0	0-1
bou	1-1	2-1	-	3-1	0-1	2-2	0-2	3-1	4-1	0-3	1-3	1-0	1-4	0-0	1-1	0-2	0-0	0-3	2-2	1-2
bha	2-1	1-1	2-0	-	1-1	1-1	0-1	3-2	0-2	1-3	0-5	0-3	0-0	2-0	0-1	0-2	3-0	1-1	1-1	2-2
bur	0-0	1-2	3-0	1-2	-	2-4	0-2	1-0	2-1	0-3	1-4	0-2	1-0	2-0	1-1	3-0	1-1	1-0	3-0	1-1
che	2-2	2-1	0-1	2-0	3-0	-	2-0	4-0	1-1	1-2	2-1	0-2	1-0	1-0	2-2	0-2	2-1	3-0	0-1	2-0
cry	1-1	1-0	1-0	1-1	0-1	2-3	-	0-0	0-2	1-2	0-2	0-2	1-0	2-0	0-1	0-2	1-1	1-0	2-1	1-1
eve	0-0	1-1	1-3	1-0	1-0	3-1	3-1	-	2-1	0-0	1-3	1-1	2-2	0-2	0-2	1-1	1-1	1-0	2-0	3-2
lei	2-0	4-0	3-1	0-0	2-1	2-2	3-0	2-1	-	0-4	0-1	0-2	5-0	1-1	2-0	1-2	2-1	2-0	4-1	0-0
liv	3-1	2-0	2-1	2-1	1-1	5-3	4-0	5-2	2-1	-	3-1	2-0	3-1	4-1	2-0	4-0	2-1	2-0	3-2	1-0
mci	3-0	3-0	2-1	4-0	5-0	2-1	2-2	2-1	3-1	4-0	-	1-2	5-0	5-0	2-0	2-1	2-2	8-0	2-0	0-2
mun	1-1	2-2	5-2	3-1	0-2	4-0	1-2	1-1	1-0	1-1	2-0	-	4-1	4-0	3-0	2-2	2-1	3-0	1-1	0-0
new	0-1	1-1	2-1	0-0	0-0	1-0	1-0	1-2	0-3	1-3	2-2	1-0	-	0-0	3-0	2-1	1-3	1-1	2-2	1-1
nor	2-2	1-5	1-0	0-1	0-2	2-3	1-1	0-1	1-0	0-1	3-2	1-3	3-1	-	1-2	0-3	2-2	0-2	0-4	1-2
shu	1-0	2-0	2-1	1-1	3-0	3-0	1-0	0-1	1-2	0-1	0-1	3-3	0-2	1-0	-	0-1	3-1	1-1	1-0	1-0
sou	0-2	2-0	1-3	1-1	1-2	1-4	1-1	1-2	0-9	1-2	1-0	1-1	0-1	2-1	3-1	-	1-0	2-1	0-1	2-3
tot	2-1	3-1	3-2	2-1	5-0	0-2	4-0	1-0	3-0	0-1	2-0	1-1	0-1	2-1	1-1	2-1	-	1-1	2-0	2-3
wat	2-2	3-0	0-0	0-3	0-3	1-2	0-0	2-3	1-1	3-0	0-4	2-0	2-1	2-1	0-0	1-3	0-0	-	1-3	2-1
whu	1-3	1-1	4-0	3-3	0-1	3-2	1-2	1-1	1-2	0-2	0-5	2-0	2-3	2-0	1-1	3-1	2-3	3-1	-	0-2
wol	0-2	2-1	1-0	0-0	1-1	2-5	2-0	3-0	0-0	1-2	3-2	1-1	1-1	3-0	1-1	1-1	1-2	2-0	2-0	-

Interpretation of the table:

- Each match is played between a home team and an away team.
- All pairs of teams play twice, switching home and away team roles.
- The first column `home_team` contains the abbreviation of home teams.
- The remaining column names are the abbreviations of away teams.
- Match results are in “home_goals–away_goals” format.
For example, “1–0” indicates that the home team won the match by one goal to zero goal.
- Winning team earns 3 points; losing team earns 0 point; both earn 1 point for a draw.

1.1 Generating the League Table (8%)

Generate a league table from the EPL match results. The table should contain the following columns from left to right:

- `rank`: From best to worst, determined by the cumulative points `points`.
 - For equal `points`, cumulative goal difference `goal_diff` is used as a tie breaker.
 - For equal `points` and `goal_diff`, `goals_scored` is used as a tie breaker.
- `team`: Full name of the team, e.g., "Manchester City."
- `points`: Cumulative points.
- `goal_diff`: Cumulative goal difference.
- `goals_scored`: Cumulative goals scored by the team.

Then, apply `gt` functions to improve the visualization of the table. For example, add a title, apply colors, adjust column names and fonts weights, apply row striping, add annotations, etc. Note: Top 4 teams qualify for Champions League, while bottom 3 are relegated.

1.2 Designing Your Own Performance Metric (12%)

In this subtask, your team will design an original per-team metric from the match results. Your analysis direction depends on your team number:

- **Even team numbers – Direction A: Home Advantage Index.**
Football fans widely believe that playing at home is an advantage. Design a metric that quantifies how much each team benefits from playing at home.
- **Odd team numbers – Direction B: Upset Index.**
An upset occurs when a supposedly weaker team takes points off a stronger one. Here, the strong teams are defined as the **top 6 teams** in the league table obtained in [Task 1.1](#). Design a metric that quantifies how much each of the remaining 14 teams over- or under-performs in its 12 matches against the strong teams. *Matches played among the 6 strong teams themselves are excluded from this analysis.*

There is no single correct answer. Follow the logic flow below for your analysis:

1. **Define** your metric precisely (with formula or computational rule) with a concise justification.
2. **Implement** the metric using `R` code, covering all 20 teams for Direction A or the 14 non-strong teams for Direction B, and present the results as a ranked `gt` table with **at least** the following columns:
 - `metric_rank`: the ranking number according to your metric
 - `team`: the full name of the team (20 for Direction A, 14 non-strong for Direction B)
 - `metric_value`: the value of your metric
3. **Verify** your proposed metric with the implementation results, with no more than 100 words.

Task 2: Investment Market Price Analysis (40%) [🔗](#)

In this task, we will analyze the historical performance of the Dow Jones Industrial Average (DJIA) and compare it with the performance of Bitcoin (BTC) and Gold. We will conduct data cleaning, visualization, and analysis to understand the trends and volatility of these assets over time.

2.1 Analyzing and Visualizing DJIA Data (15%)

Download the DJIA data `dja-performance-report-monthly.xlsx` from xSITE (**Content**→**Data**), and perform the following tasks:

1. **(2%)** Import and clean the data, create a tibble data frame named `monthly_dj`, having the following columns:
 - `date`: R's `Date` class, containing "Effective Date"
 - `close_value`: R's `numeric` class, containing "Close Value"

Display the first 10 rows of `monthly_dj`.

2. **(3%)** Identify the gap months in `monthly_dj`, e.g., one value is in January but the next value is later than February of the same year. Identify the starting point of the gap by creating a `tibble` data frame with the following columns:

- `date`
- `close_value`
- `month` (1 to 12)
- `year`
- `months_to_next_row`

Then, search online to understand the historical reason for the gap. Summarize your findings concisely (less than 50 words).

3. **(10%)** Visualizing DJIA performance by month. Append 2 columns to `monthly_dj`:
 - `rel_change_pct`: Containing the relative change of the close price (in percent).
 - `month_abb`: Containing 3-letter month abbreviations ("Jan" to "Dec") as ordered factors.

Then, visualize DJIA monthly performance using the two design methods specified below:

- A. **Combined violin and box plots** (box plots should be bounded by the violin envelope)
- B. **Faceted frequency polygons and rug plots** (monthly facets in 4 rows and 3 columns)

For Visualization A, connect successive medians across months by a line. For Visualization B, add a dashed vertical line at the mean value for each month.

2.2 Comparing DJIA, BTC, and Gold Prices (25%)

1. **(5%)** Import the BTC and gold price data since 2016 using the `tq_transmute()` function from the `tidyquant` package. The tickers are "BTC-USD" and "GC=F", respectively. Complete the following code chunk to create the `monthly_btc` and `monthly_gold` tibbles:

```
ticker_btc <- "BTC-USD"
ticker_gold <- "GC=F"
start_date <- "2016-01-01"
end_date <- "2026-06-30"

# Convert to monthly data
monthly_btc <- tq_get(ticker_btc, from = ..., to = ...) |>
  drop_na(adjusted) |>
```

```
tq_transmute(select = adjusted, mutate_fun = ..., period = ...)

monthly_gold <- tq_get(...) |>
  drop_na(adjusted) |>
  tq_transmute(select = adjusted, ...)

head(monthly_btc, 10)
head(monthly_gold, 10)
```

2. **(10%)** Create a visualization showing monthly prices of DJIA, BTC, and Gold on a single plot. The visualization-ready data frame should have the following columns (dimensions):

- **month**: From Jan 2016 to the latest month available for all assets.
- **close_value**: Close price for DJIA, adjusted close price for BTC and Gold.
- **asset**: Categorical variable indicating the asset type (DJIA, BTC, or Gold).

Note that the inconsistency of dates (if any) within the same month across the three assets can be ignored. Make your choices of the aesthetic mappings and geoms to create a clear and informative visualization. **Make sure the three price curves are visually comparable and distinguishable in your plot.**

3. **(10%)** Volatility analysis. Gold, which has been traded for thousands of years, is widely regarded as a stable asset, having served as a store of value for centuries. The DJIA, with over a century of history, represents the modern industrial economy by tracking the stock market performance of major US companies. In contrast, BTC, often described as “digital gold”, is just over a decade old and is considered a volatile asset due to its speculative behavior. Here, we will use a simple metric, namely, **the largest monthly drawdown (%)** in the past decade, to assess whether volatility increases from gold to the DJIA and to BTC. For the purposes of this assignment, drawdown is defined as the percentage decline from a peak to the subsequent trough within a given period.

Complete the **compute_drawdown** function, **verify it on the provided test vector**, and then apply it to the three price series to identify the largest drawdown for each asset:

```
compute_drawdown <- function(prices) {
  ...
}

# Verify your function before applying it to the real data:
# the largest drawdown of this test vector is 40%
prices_test <- c(100, 60, 120, 90)
compute_drawdown(prices_test)

drawdowns_gold <- ...
drawdowns_dj <- ...
drawdowns_btc <- ...

biggest_drawdown_gold <- ...
biggest_drawdown_dj <- ...
biggest_drawdown_btc <- ...

biggest_drawdown_gold
```

```
biggest_drawdown_dj  
biggest_drawdown_btc
```

What do you observe from the drawdown results? For each asset, report the **peak month** and the **trough month** of its largest drawdown, and relate these two time points to the corresponding real-world events. Do these results verify our hypothesis regarding the volatility differences among the three assets? Write a concise paragraph (no more than 100 words) to support your conclusion, referring to your computed results.

Task 3: Visualizing Singapore Tourist Spending (15%)

Download the cleaned data `singapore_tourism_receipts.csv` from xSITE (**Content**→**Data**). The data frame contains the Singapore quarterly tourism receipts by major components from 2019 to 2024, released by SingStat. It is already a pivoted longer format with the following columns:

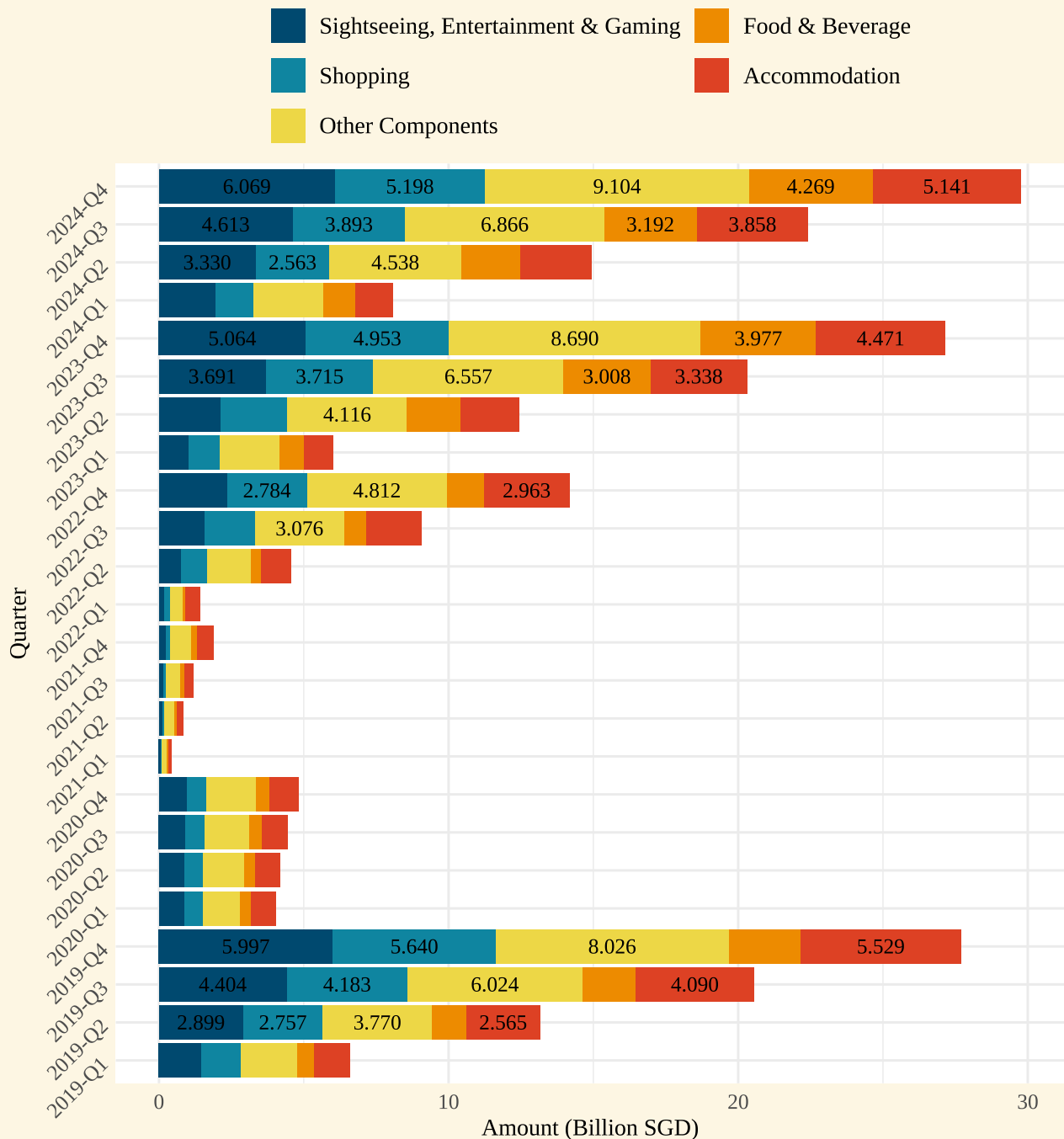
- `item`: The major components of tourism receipts including
 - Sightseeing, Entertainment & Gaming
 - Shopping
 - Other Components
 - Food & Beverage
 - Accommodation
- `quarter`: The quarter of the year in character string, "2019-Q1" to "2024-Q4"
- `value`: The receipt value in million Singapore dollar (SGD)

Task – visualizing the quarterly receipt values:

Your task here is to create a visualization showing the quarterly receipt values for each component in a single plot, following the logic flow below:

1. **Reproduce** the reference visualization shown below as closely as you can.
Use GenAI to help design scales, guides, themes, etc., which have not been covered in class.
2. **Extend** your plot with text annotations of at least two real-world events that explain major patterns in the data. Note that these events are **NOT** annotated in the reference visualization.
3. **Interpret** the visualization with no more than 100 words, making at least two specific observations that refer to actual quarters and values.

Singapore 2019–2024 Quarterly Tourism Receipts By Major Components



Task 4: Reflection on the Use of GenAI (15%)

Please note that GenAI tools are indispensable for this assessment. If you did not use any GenAI tool, you probably will not be able to complete it in time. On the other hand, if you used it properly, you may need much less time than 2 hours.

In this section, share your team's experience with using GenAI specifically for this assessment. You can discuss the following points:

- What GenAI tool did you use to help with the assessment?
- How did the tool help you? Provide concrete examples.
- What do you think is challenging about the tool?
- Any other thoughts or feedback on the use of GenAI in this assessment.

Try to use `markdown` syntax (e.g., list, hyperlink, screenshot, etc.) to format this section. Please be honest and specific in your writing and share your genuine thoughts.

 Attention

This task carries **15%** of the total marks. Superficial or generic responses will be penalized. Grading will be based on the sufficiency of content and the genuineness of your reflection.

Formatting and Submission (10%)

Submit a single file `team_XX.zip` containing:

1. `team_XX.qmd`.
2. `team_XX.html`.

Do not submit any other files or folders. Your submitted `HTML` report should be self-contained, logically structured, visually appealing, and also concise. It should display all text, code, and results, with proper styling for readability. Sections and subsections should have proper titles. Your submitted `QMD` source code should be clean and well-organized.